

Jize Wang | PhD Student at SJTU

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Education

Shanghai Jiao Tong University

Automation, advised by Prof. Cailian Chen and Prof. Xinyi Le

Research area: LLM-based agents, multi-agent systems, information extraction.

Ph.D.

2024 – 2028

Shanghai Jiao Tong University

Automation, advised by Prof. Xinyi Le

Rank: 16/52, GPA: 3.76/4.0

M.Eng.

2022 – 2024

Shanghai Jiao Tong University

Automation

Rank: 20/115, GPA: 3.73/4.0, CET-4: 627, CET-6: 550

B.Eng.

2018 – 2022

Publications

GTA: A Benchmark for General Tool Agents

Advances in Neural Information Processing Systems (NeurIPS 2024)

Jize Wang, Zerun Ma, Yining Li, Songyang Zhang, Cailian Chen, Kai Chen, Xinyi Le.

Adaptive Hinge Balance Loss for Document-Level Relation Extraction

The 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)

Jize Wang, Xinyi Le, Xiaodi Peng, Cailian Chen.

SAIL: Sample-Centric In-Context Learning for Document Information Extraction

Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2025, Oral)

Jinyu Zhang, Zhiyuan You, Jize Wang, Xinyi Le.

Data Whisperer: Efficient Data Selection for Task-Specific LLM Fine-Tuning via Few-Shot In-Context Learning

The 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)

Shaobo Wang, Xiangqi Jin, Ziming Wang, Jize Wang, Jiajun Zhang, Kaixin Li, Zichen Wen, Zhong Li, Conghui He, Xuming Hu, Linfeng Zhang.

Code Generation from Natural Language Using Two-Way Pre-Training

The 15th Conference on Advanced Computational Intelligence (ICACI 2023)

Jize Wang, Yunfeng Peng, Xinyi Le, Cailian Chen, Xinping Guan.

Research Experience

Efficient Architecture for Multi-Agent Collaboration

Shanghai Jiao Tong University, Shanghai AI Laboratory

Sep 2024 – May 2025

To address the problem of resource-intensive and propagation of hallucinations in classical Mixture-of-Agents, we design a lightweight multi-agent collaboration architecture. The work is submitted to NeurIPS 2025 (first author).

- Select the appropriate model dynamically based on the characteristics of the query, and integrate the performance and cost of each model.
- Use a sparse inference process to reduce costs, shorten inference time, and improve performance.

Tool-Use Evaluation for LLM-based Agents

Shanghai Jiao Tong University, Shanghai AI Laboratory

Oct 2023 – Jun 2024

To address the evident gaps between existing tool evaluations and real-world scenarios, we propose a benchmark to evaluate the tool-use capabilities for LLM-based agents in complex real-world scenarios. The work is accepted by NeurIPS 2024 Dataset & Benchmark Track (first author).

- Evaluations match real-world scenarios with multimodal inputs, necessitating the use of multiple tools across perception, operation, logic, and creativity for multi-step problem-solving.
- A total of 229 tasks are designed to evaluate 16 LLMs, revealing the bottlenecks in the tool-use capabilities of current LLMs in real-world scenarios.

Information Extraction from Visually Rich Documents Based on ICL

Shanghai Jiao Tong University Feb 2024 – Jun 2024
Design a fine-grained In-Context Learning (ICL) method based on entity similarity, document similarity, and layout similarity to address the challenges in layout and textual understanding, as well as the difficulty of providing accurate prompts for LLMs in information extraction from visually rich documents. The work is accepted by AAAI 2025 (oral, third author).

- Introduce fine-grained entity-level text similarity and combine it with layout similarity to enhance layout analysis of visually rich documents.
- Develop a unified ICL prompting template, enabling personalized prompts for each sample.
- Achieve state-of-the-art (SOTA) performance on mainstream datasets such as FUNSD, CORD, and SROIE.

Document-level Relation Extraction

Shanghai Jiao Tong University Oct 2022 – Jun 2023
Utilize transformer to predict relations between entities in documents. Tackle the issue of sparse positive samples in document-level extraction with a hinge-based adaptive threshold. The work is accepted by EMNLP 2023 (first author).

- Design Separate Adaptive Threshold (SAT) for multi-label classification of relations with adaptive threshold.
- Propose the adaptive hinge balance loss to address the issue of imbalance between positive and negative samples in document-level relation extraction.

Industry Experience

Research Intern, Wechat, Tencent

Research on LLM-based multi-agent systems. May 2025 – Now
○ Explore real-world scenarios within Tencent, including code engineering, data analysis, and vertical-domain writing, to optimize the performance of multi-agent systems.
○ Enhance agents’ planning, reasoning, and task-completion capabilities through techniques such as retrieval augmentation, fine-tuning, data synthesis, and reinforcement learning.

Research Intern, Shanghai AI Laboratory

Research on LLM applications. Oct 2023 – Mar 2025
Research on LLM algorithms and related applications, including LLM agents, instruction tuning, etc.
○ Develop an evaluation framework for tool agents, encompassing dataset construction, agent architecture design, tool deployment, and evaluation procedure design for popular LLMs.
○ Research on data construction algorithms for InternLM3 instruction fine-tuning, including data construction method based on Mixture-of-Agents, and an efficient MoA architecture.

Research Intern, Hikvision Shanghai

Research on visual algorithms for license plate recognition. Jul 2021
Focuses on data cleansing for new license plate recognition datasets, implementing and evaluating data cleaning methodologies.

Selected Honors and Awards

Tencent Rhino-Bird Elite Talent Program	
Tencent Inc.	2025
Agilent Special Scholarship	
Agilent Technologies Inc., Shanghai Jiao Tong University	2023
Outstanding Graduate Student	
Shanghai Jiao Tong University	2022
Zhiyuan Honor Scholarship	
Shanghai Jiao Tong University	2018 – 2021